WILLIAM GREENWOOD

SENIOR SOFTWARE ENGINEER | FULL STACK DEVELOPER
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PROFESSIONAL SUMMARY

Senior Backend & Full-Stack Engineer with over 11 years of experience architecting high-performance distributed systems, robust cloud infrastructure, and modern full stack web applications.

Proven track record of scaling enterprise backend architectures using Ruby on Rails, Go, Java, and Python while driving rapid feature iteration in startup environments via the TypeScript, Next.js, and Tailwind CSS ecosystem. Highly skilled in engineering production-ready AI workflows utilizing FastAPI and RAG architectures to automate complex operational tasks.

A product-minded technical leader who thrives in fast-paced engineering cultures, translating ambiguous product requirements into secure, scalable, and resilient technical solutions.

TECHNICAL SKILLS

Languages: Go, Python, Java, TypeScript, JavaScript, Ruby, SQL, C++, C, PHP, HTML/CSS
Backend & Frontend: Ruby on Rails, Spring Boot, FastAPI, Django, Flask, Node.js, Laravel, React, Next.js, Vue, Tailwind CSS, REST, GraphQL, gRPC
Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Kafka, schema design, query optimization, caching
Cloud & Infrastructure: AWS, Azure, GCP (Cloud Run, Cloud Functions, Cloud SQL, Pub/Sub, BigQuery, Vertex AI), Docker, Kubernetes, Terraform, Linux
AI & Automation: LLM integration, RAG applications, vector databases, prompt engineering, AI workflow automation
Engineering Practices: Microservices, event-driven architecture, CI/CD (GitHub Actions, Azure DevOps), automated testing (Jest, Cypress), observability, Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer | Stellar Science
Jan 2022 – Aug 2026 | Albuquerque, NM

- Raised system throughput 40% across enterprise backend platforms in Go, Java, and Python, enabling them to sustain high-volume workloads.
- Directed the modernization of Rails monoliths into a service-oriented architecture, extracting Go microservices behind well-defined API contracts while keeping existing consumers fully backward compatible.
- Architected fault-tolerant distributed systems on Kafka, Redis, PostgreSQL, and Kubernetes, eliminating critical bottlenecks and lifting platform uptime to 99.5%.
- Delivered highly available REST and GraphQL APIs on a gRPC service layer, hardening inter-service communication and security configuration.
- Spearheaded database tuning and cloud-native migrations across AWS and Azure, trimming API latency 12% while reducing infrastructure spend.
- Streamlined release engineering with GitHub Actions and Azure DevOps pipelines, saving roughly an hour on every deployment cycle.
- Introduced production AI workflows powered by Python, FastAPI, LLM APIs, and retrieval-augmented generation (RAG), automating manual operations and boosting operational efficiency 43%.
- Championed centralized logging, monitoring, and metrics, giving engineers faster insight during incidents and cutting mean time to resolution 17%.

Software Engineer | Noventum Custom Software
Jul 2018 – Dec 2021 | Albuquerque, NM

- Shipped full-stack features in React, Node.js, and TypeScript while shrinking frontend bundle size 40%, noticeably accelerating initial page loads.
- Implemented type-safe API communication layers that reduced client-server integration defects and cut API data overhead 60%.
- Established a modular Tailwind CSS component system with responsive layout primitives, accelerating frontend feature delivery by 17 days.

- Tuned PostgreSQL data access and Redis caching strategy, easing database load 20% during sudden traffic spikes.
- Expanded Jest and Cypress automation to 80% code coverage, measurably strengthening production stability.
- Collaborated with product owners to translate user feedback into iterative, shippable releases within rapid startup sprints.

Backend Software Engineer | Maya Virtual
Sep 2015 – May 2018 | Albuquerque, NM

- Built core backend applications from the ground up in Java, Python, and Ruby on Rails, adapting quickly to evolving business requirements.
- Modeled server-side data with strict integrity guarantees and efficient storage access patterns.
- Exposed secure REST APIs powering data exchange with several high-priority third-party partners.
- Diagnosed performance bottlenecks and resolved them through query optimization and refined data-access patterns.
- Hardened core services with enterprise-grade authentication, input validation, and proactive error handling.
- Mentored junior engineers and raised team standards through rigorous code reviews and test-driven development.

EDUCATION

Bachelor of Science in Software Engineering
New Mexico State University | Las Cruces, NM | Aug 2011 – May 2015

Project: Multi-Domain AI/ML Tactical Simulator (Feb 2022 – May 2026)
Java, Python, C++, PostgreSQL, REST APIs, Linux, Docker, CI/CD

- Engineered Java and Python backend services and REST APIs linking simulation tools, data services, and visualization applications.
- Authored data-processing pipelines for large simulation outputs and engineering datasets, pairing compute-intensive jobs with persistent storage.
- Shifted long-running computations to asynchronous processing, strengthening reliability and keeping the user experience responsive.
- sAutomated testing and deployment workflows, raising software quality and minimizing manual release effort.